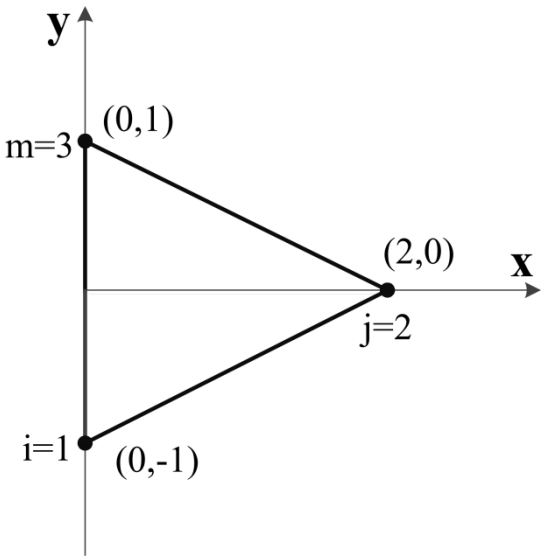
例题 1：平面单元刚度矩阵

平面应力三角形单元如图所示。结点坐标单位为英寸（in ）。

设 $E=30\times10^6$ psi，$\mu=0.25$，厚度$t=1$ in。计算单元的刚度矩阵。



解：

对于平面应力三角形单元，刚度矩阵计算列式为：

$$[K]^{e}=[B]^{T}[D][B]tA$$

其中：应变矩阵，

$$[B]=\frac{1}{2A}\begin{bmatrix} b_i & 0 & b_j & 0 & b_m & 0 \\ 0 & c_i & 0 & c_j & 0 & c_m \\ c_i & b_i & c_j & b_j & c_m & b_m \end{bmatrix}$$

本构矩阵（弹性矩阵），

$$[D]=\frac{E}{1-u^2}\begin{bmatrix} 1 & \mu & 0 \\ u & 1 & 0 \\ 0 & 0 & \dfrac{1-u}{2} \end{bmatrix}$$

由结点坐标可计算

$$b_i = y_j - y_m = 0-1 = -1$$
$$b_j = y_m - y_i = 1-(-1) = 2$$
$$b_m = y_i - y_j = -1-0 = -1$$

$$c_i = -x_j + x_m = -2+0 = -2$$
$$c_j = -x_m + x_i = 0+0 = 0$$
$$c_m = -x_i + x_j = 0+2 = 2$$

$$a_i = x_j y_m - x_m y_j = 2\times 1 - 0\times 0 = 2$$
$$a_j = x_m y_i - x_i y_m = 0\times(-1) - 0\times 1 = 0$$
$$a_m = x_i y_j - x_j y_i = 0\times 0 - 2\times(-1) = 2$$

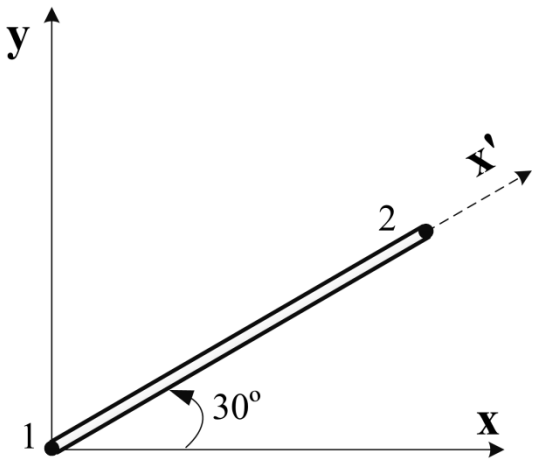
$$A = \frac{a_i + a_j + a_m}{2} = \frac{2+0+2}{2} = 2$$

将以上计算参数和给定参数代入刚度矩阵列式，得：

$$[K]^e = [B]^T [D][B] tA = 4.0\times 10^6 \begin{bmatrix} 2.5 & 1.25 & -2 & -1.5 & -0.5 & 0.25 \\ 1.25 & 4.375 & -1 & -0.75 & -0.25 & -3.625 \\ -2 & -1 & 4 & 0 & -2 & 1 \\ -1.5 & -0.75 & 0 & 1.5 & 1.5 & -0.75 \\ -0.5 & -0.25 & -2 & 1.5 & 2.5 & -1.25 \\ 0.25 & -3.625 & 1 & -0.75 & -1.25 & 4.375 \end{bmatrix} \frac{\text{lb}}{\text{in}}$$

例题 2：杆单元

图示杆单元, 横截面面积:2 in^2, 长度:60 in, 弹性模量:30×10^6 psi,
杆轴线与x 轴夹角:30°.
试计算:
1）单元在整体坐标系 x-y 中的刚度矩阵；
2）若给定单元整体坐标系结点位移u_1=0.1 in, v_1=0.0 in, u_2=0.2 in, v_2=0.3 in, 计算单元结点轴向位移、应变、应力和轴向力。



解：

杆单元刚度矩阵计算列式：

$$\left[K^{(e)}\right]=\frac{AE}{L}\begin{bmatrix} l_x^2 & l_x m_x & -l_x^2 & -l_x m_x \\ & m_x^2 & -l_x m_x & -m_x^2 \\ & & l_x^2 & l_x m_x \\ & & & m_x^2 \end{bmatrix}$$

$l_x=\cos\theta,\ m_x=\sin\theta$；$\theta$定义为$x\to x'$的转角(逆时针为正).

因此，$\theta=30^\circ\Rightarrow l_x=\cos 30^\circ=\dfrac{\sqrt{3}}{2}$、$m_x=\sin 30^\circ=\dfrac{1}{2}$

将以上计算参数和给定参数代入单元刚度矩阵计算列式，得

$$\left[K^{(e)}\right]=\frac{2\times 30\times 10^6}{60}\begin{bmatrix} \frac{3}{4} & \frac{\sqrt{3}}{4} & -\frac{3}{4} & -\frac{\sqrt{3}}{4} \\ & \frac{1}{4} & -\frac{\sqrt{3}}{4} & -\frac{1}{4} \\ & & \frac{3}{4} & \frac{\sqrt{3}}{4} \\ & & & \frac{1}{4} \end{bmatrix}\ \frac{\text{lb}}{\text{in}}$$

$$=10^6\begin{bmatrix} 0.75 & 0.433 & -0.75 & -0.433 \\ & 0.25 & -0.433 & -0.25 \\ & & 0.75 & 0.433 \\ & & & 0.25 \end{bmatrix}\ \frac{\text{lb}}{\text{in}}$$

单元结点轴向位移：

$$\begin{Bmatrix} d_1 \\ d_2 \end{Bmatrix} = [T]\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta & 0 & 0 \\ 0 & 0 & \cos\theta & \sin\theta \end{bmatrix}\begin{Bmatrix} 0.1 \\ 0.0 \\ 0.2 \\ 0.3 \end{Bmatrix}$$

$$= \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}\begin{Bmatrix} 0.1 \\ 0.0 \\ 0.2 \\ 0.3 \end{Bmatrix} = \begin{Bmatrix} 0.0866025 \\ 0.32325 \end{Bmatrix} \text{in}$$

单元轴向应变：

$$\varepsilon = \frac{du(x)}{dx} = \begin{bmatrix} -\frac{1}{L} & \frac{1}{L} \end{bmatrix}\begin{Bmatrix} d_1 \\ d_2 \end{Bmatrix} = \begin{bmatrix} -\frac{1}{L} & \frac{1}{L} \end{bmatrix}[T]\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \frac{1}{L}\begin{bmatrix} -\cos\theta & -\sin\theta & \cos\theta & \sin\theta \end{bmatrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix}$$

$$= \frac{1}{60}\begin{bmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} & \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}\begin{Bmatrix} 0.1 \\ 0.0 \\ 0.2 \\ 0.3 \end{Bmatrix} = 3.94 \times 10^{-3}$$

单元的轴向应力：

$$\sigma = E\varepsilon = \frac{E}{L}\begin{bmatrix} -\cos\theta & -\sin\theta & \cos\theta & \sin\theta \end{bmatrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix}$$

$$= \frac{30 \times 10^6}{60}\begin{bmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} & \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}\begin{Bmatrix} 0.1 \\ 0.0 \\ 0.2 \\ 0.3 \end{Bmatrix} = 1.183 \times 10^5 \text{ psi}$$

单元轴向力：

$$f = A\sigma = 2 \times 1.183 \times 10^5 = 2.366 \times 10^5 \text{ lb}$$